Worksheet 6-3 A Definite Integral Notation #1
AP Calculus Name_____

Express the limit as a definite integral on the given interval.

1.) $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{e^{x_k}}{1+x_k}\cdot\Delta x$ $[0,1]$

2.) $\lim_{n\to\infty}\sum_{k=1}^{n}[5(x_k)^3-4x_k]\cdot\Delta x$ $[2,7]$

Write a definite integral that is equivalent to the given summation notation. The lower limit for the integral is given.

3.) Integral's lower limit is 0. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\cos\left(\frac{\pi}{n}k\right)\right)\left(\frac{\pi}{n}\right)$

4.) Integral's lower limit is -3. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt[3]{-3+\frac{5k}{n}}\right)\left(\frac{5}{n}\right)$

5.) Integral's lower limit is 6. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{1}{\left(6+\frac{9k}{n}\right)^2}\right)\left(\frac{9}{n}\right)$

6.) Integral's lower limit is 0. $\lim_{n\to\infty}\sum_{k=1}^{n}\left(\tan\left(\frac{\pi}{4n}k\right)\right)\left(\frac{\pi}{4n}\right)$

7.) Integral's lower limit is -1. $\lim_{n\to\infty}\sum_{k=1}^{n}\left[4\left(-1+\frac{8k}{n}\right)\right]\left(\frac{8}{n}\right)$

Express the integral as a limit of Riemann sums.

8.) $\int_{-1}^{3} x^2 \, dx$

9.) $\int_{2}^{5} \frac{1}{x} \, dx$

10.) $\int_{3}^{4} \ln x \, dx$

11.) $\int_{1}^{3} \sqrt{4 + x^2} \, dx$

12.) $\int_{2}^{5} \left(x^2 + \frac{1}{x}\right) dx$